HR Employee Attrition Analysis

Data-Driven Insights for Workforce Stability & Retention Strategy

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Power BI | HR Analytics | Data Visualization

Why This Project Matters

The Problem

What is Attrition?

It is the rate at which employees leave a company. It shows how many people many people are quitting or moving on.

Why it hurts?

High attrition means high costs for hiring and training new people. It also lowers the team's morale and productivity.

The Solution

Data-Driven Fix

By analyzing data, we can predict who might leave and fix the problems early before they quit.

Our Objective

Build a clean Power BI dashboard to show these insights clearly so HR can take quick action.

Tools & Data Details

Tools Used

- Power BI (Visualization)
- DAX (Calculations)
- GitHub (Documentation)

Dataset Details

- 15,000 Employee Records
- Key Metrics: Satisfaction, Salary, Tenure
- Status: 1 (Left) / 0 (Stayed)

Dashboard Overview

The final interactive dashboard built to analyze attrition patterns.

Big Picture: The Key Numbers

15,000	3,570	23.8%	61.3%
TOTAL EMPLOYEES	EMPLOYEES LEFT	ATTRITION RATE	AVG SATISFACTION

What does this mean?

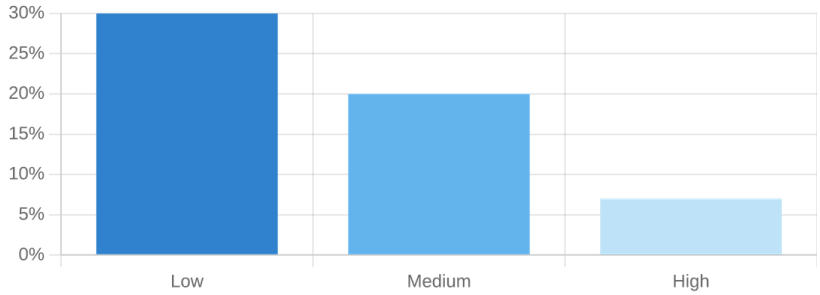
An attrition rate of **23.8%** is significant. It means nearly 1 in every 4 employees is leaving the company. This high turnover signals a need for immediate investigation into root causes to prevent further talent loss.

Insight 1: Salary & Promotion Impact

Salary Impact

Low Pay = High Risk

Employees in the low salary group show the highest attrition rate. As salary increases, the chance of leaving drops.



Promotion Impact

Lack of Growth

Employees who were not promoted in the last 5 years leave much more often than those who were.

5 Years

The critical window where lack of promotion leads to exit.

People leave when they don't feel paid well or see a future.

Insight 2: Department & Tenure Trends

Department Trends

High-Risk Areas

The **Sales, Technical, and Support** departments face the highest attrition counts.

Stable Areas

Management and R&D show the lowest attrition, reflecting better stability.

High-pressure roles need more support to keep people.

Tenure Risk Zone

The 4-5 Year Mark

Employees with **4–5 years** of experience leave the most.

Mid-Career Crisis

This is a critical stage where employees often seek new challenges or better growth elsewhere.

Mid-career employees are at the highest risk of leaving.

Insight 3: Satisfaction & Workload

Satisfaction Level

Stayed vs. Left

Employees who stay have much higher satisfaction scores (Avg 61.3%). Those who leave show a sharp drop in happiness before exiting.

Engagement

Low satisfaction is an early warning sign of "quiet quitting" or disengagement.

Workload Pattern

Monthly Hours

Employees who left often had imbalanced work hours—either extreme overwork or very low engagement (Avg 201 hours).

Burnout Risk

High-pressure departments show more irregular work patterns, leading to faster burnout.

> *"Happy employees with balanced work stay longer. Monitoring these two metrics can prevent 1 in 4 exits."*

Final Conclusion & Recommendations

The Problem Summary

Core Drivers

Low pay, limited career paths, and high-pressure teams drive the 23.8% attrition.

Critical Timing

The 4–5 year tenure is the key risk window where departures spike.

The Fix (Recommendations)

Compensation & Growth

Raise pay for low-paid roles and define clear 3-year promotion paths.

Targeted Retention

Prioritize stay interviews and wellness in the 4–5 year risk zone.

Result: A more stable, happy, and productive workforce with lower turnover costs.